Florida Qiskit Fall Fest 2026 · Team Quantum Entanglers · Project dossier for script writing · 18 Oct 2026

aperion

Imaging beyond what's visible. Quantum-assisted ultrasound brain imaging.

One-sentence pitch: A team at Imperial College London showed a helmet of ultrasound sensors can image a stroke through the skull without radiation, but one image takes 32 hours of computing; we broke their reconstruction algorithm apart, found the step that is quantum-shaped (the wave simulation, over 95% of the compute), built and tested it as a Qiskit circuit, and show honestly what a quantum computer can and cannot do for it today.

Team: Anik Sahai (clinical & product, presenter lead) · Aaron Loh (math & numerics) · Charan Jarugula (quantum).

Deck: https://claude.ai/artifact/JLxHdho2TspyzNZR7qVURJ (24 presented slides + hidden backups). For a PDF of the slides themselves, open the deck and use Share → Export → PDF. This dossier is the script source behind it.

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1. How we are judged

Criterion	Weight	What a 5 looks like	Where we hit it
Problem Definition & Relevance	15%	Compelling, specific problem; realistic scope; impact clearly argued	Slides 3, 4, 7, 8, 21–23
Quantum Rationale & Understanding	20% (tie-break)	Nuanced near-term vs future; realistic about advantage; what scale/hardware is needed	Slides 8–17, 21–22
Technical Implementation (Qiskit)	20%	Polished, reproducible; advanced techniques (error mitigation, hybrid)	Slides 16–18 (data coming)
Results & Validation	10%	Baselines, error bars/repeats, simulator vs hardware, clear conclusions	Slide 19 (data coming)
Innovation & Creativity	10%	"I hadn't thought of that"	Slides 4–6, 20
Presentation & Communication	15%	Accessible, accurate, within ~5 min	Whole deck; cover and close
Q&A, Teamwork & Learning	10%	Shared ownership, growth story, realistic roadmap	Slide 2; Q&A section below
Hardware bonus	+3 / +5	+5 = real IBM device AND noise analysis or mitigation	Needs tonight's hardware run

> **Judges are told:** “A team that clearly explains why quantum may NOT yet beat classical shows more understanding than one that overclaims.” Our honesty is a scoring strategy, not a weakness.

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2. Story arc (what the audience should feel)

1. **A huge problem:** no stroke can be treated until someone sees inside the head, and the scanners are in hospitals.
2. **A huge breakthrough:** researchers built an ultrasound helmet that sees through the skull, cheaply and without radiation.
3. **One HUGE problem:** one image takes 32 hours; every minute kills 1.9 million neurons.
4. **Our work:** we read the literature, broke their algorithm apart, found the quantum-shaped step, built it in Qiskit on top of a 1D quantum algorithm, and tested it.
5. **Honest result + future:** what works, what doesn’t yet, how it scales with quantum hardware, and the lives it could save.

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3. Slide by slide

1 Cover `Presentation` `Anik`

aperion wordmark (the “o” is a ring of transducer dots) · “Imaging beyond what’s visible.” · “Quantum-assisted ultrasound brain imaging: a clearer picture of a stroke, before the hospital.” · Team names.

> We tested whether a quantum computer can speed up the slowest step of a portable brain scanner, and we will show exactly where it works and where it does not.

2 The Team `Q&A/Teamwork` `Anik`

Three photo slots. Anik Sahai, clinical & product (Founder AuthAI; smartphone retinopathy screener; AI researcher FAU CA-AI, 2 papers submitted). Aaron Loh, math & numerics (FAU research on PDE surrogates and computer-assisted proofs; 1st nationally Mu Alpha Theta Matrices & Vectors; 2nd Florida State Science Fair). Charan Jarugula, quantum (FAU quantum research assistant on error mitigation; founder Q-Beam; computational neuroscience at UF Scripps).

> Optional chain of questions: Anik asked why stroke patients wait for scanners; Aaron asked what math takes 32 hours; Charan asked whether a quantum computer can run it and survive the noise. Each presents the part they asked about.

3 Problem

Problem 15% · Anik

Top row: **795K** US strokes/yr (11.9M worldwide) · **1.9M** neurons lost per untreated minute · **1 in 8 is a bleed** (pictogram). Bottom row: **113M** Americans lack 60-min access to clot removal; CT per million US 44.6 vs Tanzania 0.17 · **25%** of patients arrive within 3 h · **$56.2B** US cost per year.

> Picture someone collapsing at home. Paramedics arrive fast, but there are two kinds of stroke. A clot is treated with a clot-busting drug; a bleed is made worse by that same drug. So the guideline says brain imaging is essential first, and that scanner lives in a hospital. Every minute of that wait costs about 1.9 million neurons.

Sources: AHA Heart Disease & Stroke Statistics 2026 (87% ischemic, 10% intracerebral, 3% subarachnoid); GBD 2021 (Lancet Neurol 2024); Saver, Stroke 2006; 2026 AHA/ASA guideline §3.2; J NeuroIntervent Surg 2022; Stroke 2012; CDC; WHO medical devices data.

4 Competitive advantage

Innovation 10% · Anik

Category	aperion helmet	Hospital CT	Hospital MRI	CT ambulance	Portable MRI
Goes to the patient	✓	✗	✗	✓	✓
No ionizing radiation	✓	✗	✓	✗	✓
No magnet, no shielding	✓	✓	✗	✓	✗
Sees through the skull	✓	✓	✓	✓	✓
Image in minutes	✗ (32 h today)	✓	✓	✓	✓
Proven in patients	✗ (simulation)	✓	✓	✓	✓

> The helmet is the only option that goes to the patient with no radiation and no magnet. Its two gaps are time and clinical proof. Time is the one we attack. Context: CT ambulances (BEST-MSU, NEJM 2021) got treatment 36 minutes sooner (72 vs 108 min) and better outcomes (55.0% vs 44.4% little or no disability), but cost ~$1M each and cost-effectiveness is borderline; portable MRI (Hyperfine) weighs ~1,400 lb and caught 80.4% of bleeds; head CT ~1.6 mSv.

5 How it works (high level) Innovation Anik

Guasch Fig. 1 (helmet geometry and 1,024-transducer prototype) + live 2D wave simulation through a real brain slice (MNI152 template) with a speed-of-sound key and a speed profile graph.

> 1,024 transducers, each a speaker and a microphone. One fires, all listen. Sound speeds up in skull (2,814 m/s) and slows in brain (~1,505 m/s grey matter, 1,552.5 white); blood is 1,578. Those speed changes are the image. Keep the hardware high level; the interesting part is the algorithm.

6 The real breakthrough is the algorithm Innovation Anik

Reconstruction animation (blur sharpening into anatomy) + falling mismatch curve (illustrative).

> Full-waveform inversion: guess the head, simulate the sound, compare with the recordings, adjust. About 80 rounds, low frequencies first. Using the whole wave instead of just arrival time sharpens resolution from ~19 mm to ~0.9 mm. More than 95% of the work is simulating the wave.

7 “It works in simulation. But one image takes 32 hours.” Problem Anik → Aaron

Guasch Figs. 8–9: true vs reconstructed haemorrhage; 3D bleed segmentation. Big “32 h” on 128 CPU nodes × 24 cores.

> Be precise: the bleed result is a simulation, and the inversion started from the known skull. No in-vivo brain image has been published as of October 2026.

8 “Every minute kills 1.9 million neurons. A 32-hour image is no image at all.”

Problem · Quantum Aaron

Cards: 1.9M neurons/min · 4.5 h clot-busting window vs 32 h image · ~$1.2M of GPUs to get under 10 min. Banner: >95% of compute is the wave simulation.

> The device is cheap; the computer behind it is not, and the clock is brutal. The authors’ own route is 12 GPU servers (~$1.2M capital, ~$2M installed) projected to reach under 10 minutes and ~$15 to a few tens of dollars per scan at full use. That is a projection; no measured sub-10-minute 3D brain FWI has been published.

9 Literature review

Quantum Rationale · Anik/Aaron

"132 sources across stroke medicine, ultrasound imaging and quantum computing." Front pages: Guasch 2020, Cueto 2022 (Stride), Costa 2019, Schade 2024, Bösch 2025, Babbush 2023, Harrow 2009 (HHL), Nguyen 2026 (hybrid quantum-classical FWI).

> We read the paper behind the device, the software its authors use, the quantum algorithms that could replace the slow step, and the papers that limit those claims. Four research reports: clinical statistics, ultrasound/FWI, quantum algorithms, and an audit of our own code. Every number in the deck links to a source.

10 The science: a brain image is a map of how fast sound travels

Quantum Rationale · Aaron

Flow: (1) Voxels $c = (c_1 \ldots c_N)$, ~10^8 cubes of 0.5 mm · (2) Travel time $t \approx \Sigma\, \ell_i / c_i$, crossing paths triangulate · (3) Full waveform 19 mm → 0.9 mm. Guasch Fig. 2 (true · start · travel-time · FWI).

11 Overview: guess, simulate, compare, repeat (your slide)

Quantum Rationale · Aaron

1 Guess c_j · 2 Send $q(t)$ · 3 Simulate $p(x,t)$ (marked Quantum) · 4 Record d_{pred} · 5 Compare and update $E = \tfrac{1}{2}\Sigma(d_{pred} - d_{meas})^2$, $c \leftarrow c - \alpha \nabla E$, ~80 rounds. Side card: step 3 and its adjoint are >95% of compute; wave equation → Schrödinger; 29 qubits could hold a head; readout is the hard part.

12 How a computer simulates sound (derivation, 5 clicks)

Quantum Rationale · Aaron

1. Wave equation: $(1/c^2)\, \partial^2 p/\partial t^2 = \partial^2 p/\partial x^2 + q$
2. Grid: $x_j = jh$, $t_n = n\Delta t$
3. Differences: $\partial^2 p/\partial x^2 \approx (p_{j+1} - 2p_j + p_{j-1})/h^2$
4. Update rule: $p_j^{n+1} = 2p_j^n - p_j^{n-1} + (c_j^2 \Delta t^2/h^2)(p_{j+1}^n - 2p_j^n + p_{j-1}^n) + c_j^2 \Delta t^2\, q_j^n$
5. Stability: $c_{max}\, \Delta t/h \le 1$ (1D), $\le 1/\sqrt{3}$ (3D); skull sets the clock

13 How the image is found (derivation, 5 clicks)

Quantum Rationale · Aaron

1. Residual $r = F(c) - d$ (predicted minus measured)
2. Error $E = \tfrac{1}{2}\|r\|^2$ (one number)
3. Gradient $\nabla E = J_F^{\mathsf{T}} r$, computed with one extra backward ("adjoint") simulation
4. Update $c_{k+1} = \Pi[c_k - \eta(\nabla E + \lambda \nabla R)]$
5. Cost $\approx O(I \cdot S \cdot N_t \cdot V) \approx 80 \times 100 \times 2 \times 2400 \times 10^8 \approx 4 \times 10^{15}$ updates (our estimate)

> Forward = "what would I hear?" Misfit = "how wrong was I?" Adjoint = "play the error backwards to see where it came from." E = 0 does not prove the anatomy is right (non-convex, local minima).

14 Breaking their algorithm apart

Quantum Rationale · Aaron

5-box flow: Guess (classical) → Forward sim (quantum candidate) → Misfit E (quantum candidate, one number) → Adjoint sim (quantum candidate) → Update (classical). >95% of compute in the two wave simulations; anything needing a full vector stays classical.

15 Why quantum: sound becomes a Schrödinger equation (5 clicks)

Quantum Rationale · Charan

1. Grid wave equation: $d^2\varphi/dt^2 = -B\,B^{\dagger}\,\varphi$ (B = which cells touch, weighted by speed)
2. Stack $\psi = (\varphi, -i\,B^{\dagger}\varphi)$: velocity and stretch
3. $i\,d\psi/dt = H\psi$, $H = [[0, B],[B^{\dagger}, 0]]$, the equation a quantum computer runs
4. $\|\psi\|^2$ = wave energy, conserved = probability conserved
5. $\psi(t) = e^{-iHt}\psi(0) \approx (e^{-iH_{even}\,\Delta t}\, e^{-iH_{odd}\,\Delta t})^n\,\psi(0)$: Trotter steps

> Plain English: the wave becomes a packet hopping along a chain of cells and links. n qubits hold 2^n amplitudes, so 29 qubits could hold a whole head (~341M sites). But you can only read it out one sample at a time. Realistic gain is polynomial, at best quartic in 3D (Bösch 2025), not exponential. Costa's H carries a 1/a factor, absorbed into B here.

16 The 1D quantum algorithm we build on (Schade et al. 2024)

Quantum Rationale · Charan

Five steps: discretize the 1D elastic wave (varying ρ, μ) → rewrite as Schrödinger → load 2^n grid points into n qubits → evolve with e^{-iHt} → measure. Their Qiskit circuit (state prep · time evolution · observable · measure). "64 grid points → 6 qubits; a head's 341M sites → 29."

17 It runs on a real IBM quantum computer, at toy scale

Quantum Rationale · Charan

Their figures: 64-point simulator matches the classical ODE solver; 9-point run on IBM Brisbane (blue) drifts from classical (black). Stats: 4–5 qubits · ~870 gates, no error mitigation · up to 60% error. Our addition: tissue speeds, Trotter steps, error mitigation, skull and bleed.

> Note: Schade compiled the exact unitary (not Trotterized) and read out by state tomography at 1,000 shots. IBM Brisbane was retired in Nov 2025.

18–20 Technical steps & tests · Results · Innovation

Technical · Results · Innovation

Left blank for the team's experiment data. See section 5 for what the code already shows and what to run tonight.

21 Conclusion · Lives saved

Problem · Quantum · Anik

36 min faster treatment when imaging comes to the patient (BEST-MSU) · ~68M neurons in those 36 min (our calc: 36 × 1.9M) · 49M Americans 60+ min from advanced stroke care (CDC 2025). Scaling: today 15 noisy qubits on one line through the head → IBM Starling 2029 targets 200 error-corrected qubits and 10^8 operations → a head needs 29 logical qubits and ~10^8+ reliable operations, asked for one number at a time. Must be true: error-corrected qubits, efficient loading, one-number readout.

22 Conclusion · Making the technique useful

Problem · Quantum · Anik

Path: Today 32 h CPU → GPUs ~10 min (projected) → hybrid quantum for the wave step via the cloud → every clinic (helmet + cloud compute). Unlocks: triage in the ambulance; repeat scans (no radiation, so a bleed can be watched hour by hour); reach where CT never does (0.17 scanners per million).

> The technique only becomes useful when the image is fast and cheap. Quantum is the last step on that path, not the first.

23 **Conclusion · Imaging beyond what's visible** Presentation Anik

Uses: stroke in the ambulance; low-income clinics; breast and limb (ultrasound tomography already FDA-approved for breast: Delphinus SoftVue, 2021); beyond medicine (seismic, non-destructive testing). Belief: "We believe the wall is compute, not physics. Classical hardware will get there first; quantum is how the same scan could reach every clinic on the cloud. Every minute we save is 1.9 million neurons." Theme tie-in: "A decade of quantum on the cloud."

24 **References** (backup)

4. The ideas in plain English (for the script)

Ultrasound tomography / full-waveform inversion

- The image is a list of sound speeds in ~10^8 tiny cubes (0.5 mm). Sound is faster in skull, slower in brain, slightly faster in blood than brain (~5% vs grey matter, ~1.7% vs white).
- FWI guesses that list, simulates the sound through it, compares predicted and measured recordings, and nudges the guess. Analogy: guessing what's in a dark room by clapping and listening to echoes.
- Each round needs a forward and an adjoint wave simulation for every transmitter: that is >95% of the compute and why it took 32 h.

Why the wave fits a quantum computer

- Quantum computers only rotate a vector while keeping its length fixed. Lossless sound keeps total energy fixed. Build a vector whose length is the energy, and the wave becomes a quantum evolution.
- Picture: a packet of amplitude hopping along a chain of cells and the links between them; hops are faster through skull. Our 7 cells + 8 links = 15 sites.
- **Two encodings:** one qubit per site (15 qubits, cheap neighbour gates, 238 two-qubit gates, but only 15 numbers in play, so a laptop can do it) vs compact binary (4 qubits for 15 sites; n qubits hold 2^n amplitudes, 29 for a head; but neighbours aren't neighbouring bits, so 1,414 gates for 7 cells).
- **Readout wall:** each shot reveals one site with probability = its energy share. Reading a full FWI round (~10^9 numbers) would take ~10^{13} shots (~114 years at IBM's rate). So ask for one number only (misfit, or bleed yes/no). Amplitude estimation could cut shots from $1/\varepsilon^2$ to ~$1/\varepsilon$ but needs deep circuits.
- **Time steps don't shrink:** sound still crosses ~440 cells, so quantum saves memory per step, not the number of steps.

5. Our code and results (team-quantum-ultrasound repo)

Item	Result	Status
7-cell line	scalp 1600 · skull 2800 · brain 1546 · brain 1546 (bleed 1578) · brain 1546 · skull 2800 · scalp 1600 m/s	—
Hamiltonian	$H = [[0, iK], [-iK^T, 0]]$, $K = \mathrm{diag}(c)\cdot B$ (reproduces $p_{tt} = c^2\nabla^2 p$)	Done
Classical solver vs Stride (researchers' code)	0.993 mean trace correlation	Measured
Two classical solvers (3D 7^3)	agree to 7×10^{-8}; energy drift 4×10^{-6}	Measured
Our 2D FWI	bleed recovered 1,576 vs 1,578 m/s; brain 1,546 vs 1,546; misfit $1.9\times10^{-5} \rightarrow 1.3\times10^{-6}$ in 40 rounds	Measured
Exact target	energy at first brain cell, t = 1.5: 0.1040 (bleed 0.1041)	—
Unary (one per site)	15 qubits, 238 CZ, depth 130 (Kingston layout); noiseless error 2.4% on signal-bearing cell at 8 steps	Simulator
Compact	4 qubits, 336/694/1,414 CZ at 2/4/8 steps	Simulator
2D 3×3	33 / 6 qubits; 882 / 36,416 CZ	Simulator
3D 3×3×3	135 qubits, error 0.0057, 4,374 CZ (grid too small to contain brain)	MPS simulator
IBM noise model (FakeKingston), 8 steps, 4,000 shots, 1 seed	raw 0.047 ± 0.003 · post-selected 0.096 ± 0.007 (48.8% kept) · + ZNE 0.103 ± 0.011	Noise model
Full head 440^3	341M sites → 29 compact qubits; unary 3.4×10^8 qubits, $\geq2\times10^{13}$ CZ (extrapolated)	Estimate
Hybrid one-number misfit (7^3, emulated)	correct bleed location: 50% at 10^{10} shots, 95% at 10^{11}	Emulated
Timing	laptop exact solve 23 µs, finite difference 1.8 ms; QPU ~3 s per 4,000-shot circuit (estimate)	Measured / estimate
Real IBM hardware	not yet run (RUN_ON_IBM = False)	**Tonight**

To run tonight (≈30–60 s of the 10-min Open Plan quota)

1. Set RUN_ON_IBM = True; run the 6 circuits (2/4/6/8 steps, fold-3 at 8, bleed at 8) × 3 repeats; record backend name and job IDs.
2. Add healthy vs bleed at t = 5.5 on the bleed cell (~20k shots each): bleed effect there is −0.014, ~300× larger than at the current receiver (4.7×10^{-5}).
3. Repeat noise-model runs over 5 seeds for real error bars; add a third ZNE noise level.

4. Screenshot pytest (35 tests) and draw one Trotter step of the circuit.
5. Be ready to explain 06_stride_validation.png (no plotting script in repo; its “1.000” bar is 0.9997).

Qiskit notes: SamplerV2 supports twirling and dynamical decoupling; TREX/ZNE/PEA are EstimatorV2 options. PEA needs an extra noise-learning job that uses quota.

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6. Claims: say this, not that

Avoid	Say instead
“Published in Nature”	“Published in npj Digital Medicine, a Nature Portfolio journal”
“Quantum reconstructs brain images”	“We turned the slowest step into a quantum circuit, ran it, and measured what a full head would need”
“Exponential speedup”	“Memory saving is exponential on paper; realistic net gain is polynomial, at best quartic in 3D”
“First quantum wave simulation / first quantum FWI”	“To our knowledge, the first scalp–skull–brain wave model run on IBM hardware with error mitigation” (only after tonight’s run)
“The helmet costs $2–3K”	No source found; say “cheap sensors, expensive compute”
“A stroke every 40 seconds”	“795,000 strokes a year” (40 s is our arithmetic)
“Two-thirds of the world lacks imaging”	“CT scanners per million: US 44.6, Tanzania 0.17”
Noise-model numbers as hardware	Always label FakeKingston results “IBM noise model”
“The helmet detects bleeds in patients”	“In simulation, starting from the known skull”

7. Q&A prep

Question	Owner	Answer core
Why quantum instead of classical?	Charan	The slow step is an energy-conserving wave simulation, which maps exactly onto unitary evolution; we ruled out annealing/QAOA (optimization), VQE (ground states), QML (needs data), HHL (readout).
What would need to improve to beat classical?	Charan	Error-corrected qubits (~29 logical for a head), ≥10^8 reliable operations, efficient state loading, one-number readout.
Walk me through your circuit.	Charan	X on qubit 0 (the kick), Trotter steps of neighbour hops (even links, odd links, even), measure; probability on receiver qubit = energy there (0.104).
How do you know it worked?	Aaron	Classical baseline validated against Stride (0.993); circuit matches exact; noise model + mitigation recovers 0.103 vs 0.104.
What surprised you?	Aaron	The readout, not the simulation, is the wall; and the receiver we first chose barely sees the bleed.
Is this clinically real?	Anik	No. The helmet's bleed result is simulation; no in-vivo brain image yet. We make no clinical claim.
What did each of you do?	All	Anik: problem, product, story. Aaron: the PDE math and classical baseline. Charan: quantum circuits and error mitigation.
What would you do with another month?	Anik	Hardware runs with repeats, bleed-sensitive readout, a one-number misfit circuit, block-encoded compact circuits, absorption.

8. Key sources

1. Guasch, Calderón Agudo, Tang, Nachev & Warner. Full-waveform inversion imaging of the human brain. npj Digital Medicine 3, 28 (2020). CC BY 4.0.
2. Schade, Bösch, Hapla & Fichtner. A quantum computing concept for 1-D elastic wave simulation. Geophys. J. Int. 238 (2024); arXiv:2312.14747v2. CC BY 4.0.
3. Costa, Jordan & Ostrander. Quantum algorithm for simulating the wave equation. Phys. Rev. A 99, 012323 (2019).
4. Bösch et al. Quantum wave simulation with sources and loss functions. Phys. Rev. Research 7, 033225 (2025).
5. Cueto et al. Stride. Comput. Methods Programs Biomed. 221 (2022). Babbush et al. PRX 13 (2023). Harrow, Hassidim & Lloyd 2009. Nguyen et al. arXiv:2606.01110 (2026).
6. AHA Heart Disease & Stroke Statistics 2026; 2026 AHA/ASA AIS guideline; Saver, Stroke 37:263 (2006); GBD 2021 (Lancet Neurol 2024); CDC 2025; BEST-MSU (NEJM 2021); Hyperfine (Nat Commun 2021); WHO devices data.
7. IBM Quantum roadmap (Starling 2029, Blue Jay 2033+). Brain anatomy: MNI152 ICBM 2009 template.
8. Full source list with links and exact quotes: research/R1–R4 in the project folder (132 unique sources).